GAME THEORY & STRATEGY
A Premium University-Level Study Companion

BOOK 3
End-Term Preparation Guide

Textbook + Strategy Handbook + Complete PYQ Solutions
rolled into one self-contained resource

Coverage

Week 9: Cooperative Games & Shapley Value

Week 10: Fair Division, Bankruptcy & Cost-Sharing

Week 11: Auctions

Week 12: Network Effects & Sponsored Search

Assessment: End-Term PYQs · Full Solutions · Strategy Guide · Concepts found only in PYQs

How to Use This Book

This book is engineered to be *sufficient on its own* for scoring full marks in the End-Term. You should be able to put away the slides and lecture videos and study only from here.

- **Part I — Complete Notes (Weeks 9–12).** Every concept is built from first principles. Read the intuition boxes first if you are new, then the definitions, then the worked examples. No prior economics or mathematics is assumed.
- **Part II — PYQ Solutions.** The recurring End-Term families are solved with verified answer keys (network-effects demand, Shapley & power index, bankruptcy, auctions, VCG, Bayes), with cumulative Quiz-1/Quiz-2 items cross-referenced to Books 1–2.
- **Part III — Concepts Found Only in PYQs.** Order-based Shapley, order statistics for auctions, bankruptcy rules, VCG pricing, network demand.
- **Part IV — Quick Revision Sheets.** One-page summary, formula card, trap list.
- **Part V — Exam Strategy Guide.** Triage table and per-family playbook.

The box colour code. Definition Intuition / FAQ Why it exists Equilibrium insight Exam tip / Remember Mistake / Trap / Issue.

★ The arc of the End-Term

The End-Term is dominated by four big topics: **cooperative games & the Shapley value** (computing fair divisions), **fair division, bankruptcy & cost-sharing**, **auctions** (the four formats, truthful bidding, revenue equivalence), and **network effects & sponsored-search (VCG) markets**. The common thread is *value creation and its division* — who creates worth, and how it is shared or priced.

Contents

Part I

Complete Notes (Weeks 9–12)

CHAPTER 1

Week 9 — Cooperative Games and the Shapley Value

The question this week answers

A group of people create value *together*. How should they split it *fairly*? Quiz 2 introduced the *core* (which splits survive coalitional defection). The End-Term goes further: the **Shapley value** gives a *single*, unique, fair number for each player — "what an impartial judge would recommend." It is the most heavily tested idea in this block.

1.1 Cooperative Games Recalled

Transferable-utility cooperative game (N, v)

$N = \{1, 2, \ldots, n\}$ is the set of players. The **coalition function** (characteristic function) v assigns to every subset (*coalition*) $S \subseteq N$ a number $v(S)$, the **worth** the coalition can generate on its own, with $v(\varnothing) = 0$. "Coalition S forms" means all members of S consent. *Transferable utility* means the worth is money that can be split in any way among members. Binding agreements are enforceable.

The advertising game (running example)

An agent wants ≥ 2 of three celebrities to endorse a brand. Pay-outs: $\{1,2\} \to 100$, $\{1,3\} \to 100$, $\{2,3\} \to 50$, all three $\to 120$; any single signer earns 0.

$$v(1) = v(2) = v(3) = 0, \quad v(12) = 100,\ v(13) = 100,\ v(23) = 50,\ v(123) = 120.$$

What should an impartial judge recommend each player receive? (Answer below, once we have the Shapley value.)

Useful classes of games

- **Additive:** $v(S \cup T) = v(S) + v(T)$ for disjoint S, T — no synergy; each player is an island.
- **Superadditive:** $v(S \cup T) \geq v(S) + v(T)$ — merging never hurts, so the grand coalition N is efficient to form. (Most "value-creation" games are superadditive.)
- **Simple / voting (majority) game:** $v(S) \in \{0, 1\}$; $v(S) = 1$ iff S is "winning" (e.g. total weight $\geq$ quota q).

1.1.1 Special players

Symmetric and null players

Players i, j are **symmetric** if swapping them never changes any coalition's worth: $v(S \cup \{i\}) = v(S \cup \{j\})$ for every S containing neither. Player i is a **null (dummy) player** if he adds nothing anywhere: $v(S \cup \{i\}) = v(S)$ for every S not containing i. Symmetric players *deserve* equal shares; a null player *deserves* zero.

1.2 The Shapley Value — Axioms

Simple Intuition

Rather than guess a fair split, Shapley listed properties any fair rule *must* have, then proved exactly one rule satisfies them all. This "axiomatic" approach is why the Shapley value is so compelling — it is the *unique* fair division.

The four Shapley axioms

A value $\phi = (\phi_1, \ldots, \phi_n)$ assigns each player a payoff. The Shapley value is the *unique* ϕ satisfying:

1. **Efficiency:** $\sum_{i \in N} \phi_i = v(N)$ — the whole pie is divided.
2. **Symmetry:** symmetric players get equal payoffs.
3. **Null player:** a null player gets 0.
4. **Additivity:** for two games on the same players, $\phi(v + w) = \phi(v) + \phi(w)$ — valuing a combined venture = summing the separate valuations.

Lloyd Shapley (1953) proved these four pin down *one* value.

1.3 Computing the Shapley Value: the Random-Order Rule

The "enter the room" procedure

Imagine the players entering an empty room *one at a time in a random order*. As each enters, she is paid her **marginal contribution** — the extra worth she adds to those already inside. Average each player's marginal contribution over *all $n!$ orders* (each equally likely). **That average is the Shapley value.**

$$\boxed{\phi_i(v) = \frac{1}{n!} \sum_{\text{orders}} \big[v(\text{predecessors} + i) - v(\text{predecessors})\big]}$$

Equivalently, summing over coalitions S not containing i:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \big[v(S \cup \{i\}) - v(S)\big].$$

Full computation — Example 18

$N = \{1, 2, 3\}$ with $v(1) = 6, v(2) = 12, v(3) = 18,\ v(12) = 30, v(13) = 60, v(23) = 90,\ v(123) = 120$. There are $3! = 6$ orders. For each, pay each entrant her marginal contribution:

Order	ϕ_1	ϕ_2	ϕ_3
123	6	30−6=24	120−30=90
132	6	120−60=60	60−6=54
213	30−12=18	12	120−30=90
231	120−90=30	12	90−12=78
312	60−18=42	120−60=60	18
321	120−90=30	90−18=72	18
Sum	132	240	348
÷6	**22**	**40**	**58**

$\boxed{\phi = (22,\ 40,\ 58)}$, and indeed $22 + 40 + 58 = 120 = v(N)$ ✓ (efficiency).

Two quick cases you should be able to do instantly

Two-player bargaining $v(1) = v(2) = 0$, $v(12) = 1$: by symmetry each adds $\frac{1}{2}$ on average, $\phi = (\frac{1}{2}, \frac{1}{2})$. **Buyer–seller** $v(1) = 100{,}000$ (seller's reservation), $v(2) = 0$, $v(12) = 150{,}000$: orders 12 and 21 give seller $\{100k,\ 150k\}$ and buyer $\{50k,\ 0\}$; averages $\phi_{\text{seller}} = 125{,}000$, $\phi_{\text{buyer}} = 25{,}000$ — the \$50k gain from trade is split equally on top of the seller's reservation.

Using the axioms as shortcuts (4-player example)

$v(1) = v(2) = 100$, $v(3) = v(4) = 0$; $v(12) = 200$, $v(34) = 0$, all mixed pairs $= 150$; $v(123) = v(124) = 250$, $v(134) = v(234) = 150$, $v(1234) = 300$. Players $1, 2$ are symmetric, and $3, 4$ are symmetric. Computing player 1's average marginal contribution over the $4! = 24$ orders gives $\frac{3000}{24} = 125$; by symmetry $\phi_2 = 125$. Efficiency then forces $\phi_3 + \phi_4 = 300 - 250 = 50$, and symmetry splits it: $\phi_3 = \phi_4 = 25$. So $\phi = (125, 125, 25, 25)$. **Lesson:** use symmetry/null/efficiency to avoid enumerating all orders.

Answer to the advertising game

For $v(12) = v(13) = 100$, $v(23) = 50$, $v(123) = 120$ the Shapley value is

$$\phi = \left(\tfrac{170}{3},\ \tfrac{95}{3},\ \tfrac{95}{3}\right) \approx (56.67,\ 31.67,\ 31.67).$$

Player 1 (wanted by *both* partners) earns more than the symmetric pair $2, 3$; the three shares sum to $120 = v(N)$. An impartial judge would recommend exactly this split.

Common Mistake

Marginal contribution depends on who is already in the room, so you must track the *predecessor set*, not just the player. A frequent error is averaging $v(\{i\})$ instead of $v(S \cup \{i\}) - v(S)$. Also: the weights $\frac{|S|!(n-|S|-1)!}{n!}$ are *not* all equal — coalitions of different sizes get different weights (there are more orders that produce a small or large predecessor set).

★ How Shapley questions appear

Usually: "given this v, find the Shapley value of player X." Fastest route for $n = 3$: list the 6 orders (or use $\phi_i = \frac{1}{6}\big[2v(i) + (\text{two terms with one partner}) + 2(v(N) - v(N \setminus i))\big]$).

For $n = 3$ explicitly,

$$\phi_i = \tfrac{1}{3}v(i) + \tfrac{1}{6}\big[(v(ij) - v(j)) + (v(ik) - v(k))\big] + \tfrac{1}{3}\big(v(N) - v(jk)\big).$$

Memorise this $n = 3$ formula — it answers most exam Shapley items in one line.

1.4 Voting Power: the Shapley–Shubik Index

Simple Intuition

In a committee, "power" is not the same as "number of votes." The Shapley value of a *majority game* measures how often a member is **pivotal** — the one whose joining turns a losing coalition into a winning one. Applied to voting, the Shapley value is called the **Shapley–Shubik power index**.

Majority / weighted-voting game

Members have weights $w_1, \ldots, w_n$ and a quota q. A coalition S is *winning* ($v(S) = 1$) if $\sum_{i \in S} w_i \geq q$, else $v(S) = 0$. The Shapley value then equals the fraction of the $n!$ orders in which member i is the *pivot* (the entrant who first pushes the running total to $\geq q$).

Three-member committee, majority rule

$v(S) = 1$ iff $|S| \geq 2$ (any two of three pass a decision). By symmetry each member is pivotal in exactly $\frac{1}{3}$ of orders, so the power index is $\phi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ — equal power, as it should be. But change the weights, say $w = (50, 49, 1)$ with quota 51: now *every* winning two-member coalition still needs exactly two members, so all three remain symmetric and power is again $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ — the tiny third member has the *same* power as the giant! Voting power $\neq$ vote share: this counter-intuitive result is a favourite exam point.

1.5 Carrier, and Shapley vs. the Core

Carrier

A coalition T is a **carrier** if $v(S) = v(S \cap T)$ for every S — all the "power" lives in T, and players outside T are null. The Shapley value automatically pays outsiders 0 and divides $v(N)$ among the carrier.

Two different "fairness" answers

The **core** (Week 6) gives the set of *stable* divisions (no coalition can profitably defect); it can be empty or a whole range. The **Shapley value** gives a single *fair* division (average marginal contribution); it always exists and is unique. They need not coincide: the Shapley value can lie *outside* the core, and is guaranteed to lie *inside* it only for **convex** games (where $v(S \cup \{i\}) - v(S)$ grows as S grows). Use the core for "what splits are stable?" and the Shapley value for "what split is fair?"

	Core	Shapley value
Answers	Which splits are *stable*?	What split is *fair*?
Output	A set (maybe empty / a range)	A unique point
Defined by	$\sum_S x_i \geq v(S)\ \forall S$	Four axioms / avg. marginal contribution
Existence	Not guaranteed	Always exists
Relationship		In the core iff the game is convex

1.6 Why the Shapley Weights Look Like That

Deriving the coefficient $\frac{|S|!\,(n-|S|-1)!}{n!}$

The order rule says "average the marginal contribution over all $n!$ orders." Group the orders by *which set S sits in front of player i*. For a fixed predecessor set S (with $|S| = s$), the number of orders in which exactly those s players come before i and the remaining $n - s - 1$ come after is

$$\underbrace{s!}_{\text{arrange } S \text{ before } i} \times \underbrace{(n-s-1)!}_{\text{arrange the rest after } i}.$$

Dividing by the total $n!$ gives the weight $\frac{s!\,(n-s-1)!}{n!}$ attached to the marginal contribution $v(S \cup \{i\}) - v(S)$. So the "unequal" weights are simply *counting how many orders* produce each predecessor set — middle-sized coalitions are rarer than very small or very large ones, which is why their weights differ.

1.7 More Worked Examples

The glove market (a sharp Shapley result)

Players $1, 2$ own a *left* glove each; player 3 owns a *right* glove. A matched pair sells for \$1; a single glove is worthless. So

$$v(13) = v(23) = 1,\ v(123) = 1,\ \text{all else } 0.$$

Compute ϕ_3 (the right-glove owner) by the 6 orders — player 3 is *pivotal* (adds 1) whenever he arrives *after* at least one left-glove owner:

Order	MC_1	MC_2	MC_3
123	0	0	1
132	0	0	1
213	0	0	1
231	0	0	1
312	1	0	0
321	0	1	0
$\div 6$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{4}{6}$

$$\boxed{\phi = \left(\tfrac{1}{6}, \tfrac{1}{6}, \tfrac{2}{3}\right).}$$

The *scarce* side (the lone right glove) captures two-thirds; the two left-glove owners, competing against each other, get only $\frac{1}{6}$ apiece. **Scarcity is power** — the same lesson as the seller/buyers core in Week 6, now as a single fair number.

Same game, two methods agree — a 3-player check

Take $v(1) = 0, v(2) = 0, v(3) = 0,\ v(12) = 4, v(13) = 2, v(23) = 6,\ v(123) = 9$. **Method 1 (the $n = 3$ formula) for player 1:**

$$\phi_1 = \tfrac{0}{3} + \tfrac{(v(12)-v(2))+(v(13)-v(3))}{6} + \tfrac{v(123)-v(23)}{3} = \tfrac{(4-0)+(2-0)}{6} + \tfrac{9-6}{3} = \tfrac{6}{6} + 1 = 2.$$

Similarly $\phi_2 = \frac{(4)+(6-0)}{6} \cdots$: $\phi_2 = \frac{(v(12)-v(1))+(v(23)-v(3))}{6} + \frac{v(123)-v(13)}{3} = \frac{4+6}{6} + \frac{7}{3} = \frac{10}{6} + \frac{7}{3} = \frac{10+14}{6} = 4$. And $\phi_3 = \frac{(2)+(6)}{6} + \frac{9-4}{3} = \frac{8}{6} + \frac{5}{3} = \frac{8+10}{6} = 3$. **Method 2 (efficiency check):** $2 + 4 + 3 = 9 = v(123)$ ✓. Two routes, one answer $\phi = (2, 4, 3)$.

A power index with *unequal* power: $[3; 2, 1, 1]$

Quota 3; weights $(2, 1, 1)$. Winning coalitions: $\{1, 2\}, \{1, 3\}$ (each $= 3$), $\{2, 3\} = 2$ *losing*, all triples winning; singletons losing. Count pivots over the $3! = 6$ orders (the pivot is whoever first pushes the running weight to ≥ 3):

$$123{:}\,2,\ \ 132{:}\,3,\ \ 213{:}\,1,\ \ 231{:}\,1,\ \ 312{:}\,1,\ \ 321{:}\,1$$

(reading the pivotal player in each order): player 1 is pivotal in 4 orders, players 2 and 3 in 1 each. Power index $\left(\frac{4}{6}, \frac{1}{6}, \frac{1}{6}\right) = \left(\frac{2}{3}, \frac{1}{6}, \frac{1}{6}\right)$. **The big member has $4\times$ the power of each small member** — unlike the $[61; 40, 40, 30, 10]$ game where the small member was *null*. Power depends on the *pivot structure*, not the raw weights.

Reading a power index

A player with **veto** power (in every winning coalition) has the largest index; a **null** player (in no pivotal position) has index 0; **symmetric** members share equally. Always sanity-check that the indices sum to 1.

End of Week 9

Quick Revision — Week 9

Cooperative game (N, v): worth of each coalition. **Shapley value** = unique fair division satisfying *efficiency, symmetry, null-player, additivity*. Compute it as the **average marginal contribution over all $n!$ entry orders**. Symmetric players share equally; null players get 0; shares always sum to $v(N)$.

Formula Sheet — Week 9

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \big[v(S \cup i) - v(S)\big]; \qquad \phi_i^{(n=3)} = \tfrac{v(i)}{3} + \tfrac{(v(ij)-v(j))+(v(ik)-v(k))}{6} + \tfrac{v(N)-v(jk)}{3}.$$

★ Algorithm sheet — Shapley by hand ($n = 3$)

(1) List the 6 orders. (2) For each, compute each player's marginal contribution against her predecessors. (3) Sum each column and divide by 6. (4) Check the three add to $v(N)$. Use symmetry/null shortcuts whenever players are interchangeable or contribute nothing.

Practice questions.

1. Find the Shapley value of the advertising game and confirm $\phi = (\frac{170}{3}, \frac{95}{3}, \frac{95}{3})$.
2. Majority game $v(S) = 1$ iff $|S| \geq 2$ (three players): show $\phi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.
3. In a game with a null player, prove the null player's Shapley value is 0 directly from the order rule.
4. Two-seller/one-buyer market $v(13) = v(23) = 200,\ v(12) = 200,\ v(123) = 300$ (others as in lecture): compute ϕ.
5. Explain why additivity lets you value a merged project as the sum of separate Shapley values.

CHAPTER 2

Week 10 — Fair Division, Bankruptcy and Cost-Sharing

Same tool, three guises

Week 10 applies cooperative-game ideas to concrete division problems: splitting the proceeds when partners *dissolve* a firm, sharing the *cost* of shared infrastructure, and dividing too-few *assets* among creditors in *bankruptcy*. A recurring punchline: the "fair" answer is often the Shapley value in disguise.

2.1 Dissolving a Partnership

Simple Intuition

Partners jointly own assets and decide to sell up. Splitting in proportion to investment is one option — but it ignores how much each partner's assets are worth *in combination*. A fairer rule uses a *claims* procedure.

The claims procedure

Coalitions "claim their worth" in increasing size: first each single player claims $v(\{i\})$ (taking back what he brought), then each pair claims its worth, finally the grand coalition. Each time a coalition claims, its worth is *divided equally among its members*, and that worth is subtracted from every larger coalition containing it.

Result 1

The dissolving procedure **always terminates**, and the final division of proceeds is exactly the **Shapley value** of the game — *independent of the order* in which coalitions claim. So "everyone takes back what they brought, and the joint surplus is shared equally at each level" *is* the Shapley value. This is the cleanest intuition for what the Shapley value *does*.

The garage–petrol–restaurant partnership

Three partners own assets worth, separately, $v(1) = 30$ (garage), $v(2) = 12$ (petrol pump), $v(3) = 6$ (restaurant). Combinations: $v(12) = 36$ (they jointly bought a repair shop), $v(13) = 36$, $v(23) = 30$, and $v(123) = 90$ (the showroom they all built). They sell everything (total 90) and divide.

Claims procedure (one order): player 3 claims the restaurant, takes 6; subtract 6 from every coalition containing 3, giving a reduced game $u(1) = 30, u(2) = 12, u(3) = 0, u(12) =$

$36, u(13) = 30, u(23) = 24, u(123) = 84$. Continue with the pairs and grand coalition; the surplus at each level is split equally among that coalition's members. Carrying this out (or averaging marginal contributions over all 6 orders) gives

$$\boxed{\phi = (39,\ 27,\ 24)}, \qquad 39 + 27 + 24 = 90\ \checkmark.$$

Player 1 (the garage, most valuable standalone *and* a useful partner) gets the most; the order of claims does not matter — every order yields $(39, 27, 24)$.

2.2 Cost-Sharing Games

Simple Intuition

Several users share a piece of infrastructure (a cable network, a runway, a road). How should the *total cost* be split? Model it as a cooperative game where $v(S)$ = the cost coalition S would bear *on its own*, and apply the Shapley value.

A TV-cable network (Shapley = equal split per arc)

A station connects three customers along a tree of cables. The cost of each cable "arc" is shared, and a customer "uses" an arc if it lies on the path connecting her to the station. Suppose the arc costs and usage are such that the total is 60, with: a shared trunk arc (cost 6) used by *all three*; an arc (cost 12) used only by player 2; an arc (cost 18) used by players 1 and 3; an arc (cost 24) used only by player 3.
Shapley rule for tree-cost games: split each arc's cost *equally among the players who use it.*

$$\phi_1 = \tfrac{6}{3}+\tfrac{18}{2} = 2+9 = 11, \quad \phi_2 = \tfrac{6}{3}+12 = 2+12 = 14, \quad \phi_3 = \tfrac{6}{3}+\tfrac{18}{2}+24 = 2+9+24 = 35.$$

$$\boxed{\phi = (11,\ 14,\ 35)}, \qquad 11 + 14 + 35 = 60\ \checkmark.$$

For such network/airport problems the Shapley value coincides with this transparent "equal-split-per-arc" procedure — a result worth memorising.

Stand-alone (core) constraints

Cost allocations must also respect *standalone* limits: a community will not pay more than it would cost to build alone. E.g. if two communities can each build for 11 and 7 but jointly for 15, an equal split (7.5 each) is rejected by community B (it can do it alone for 7); a core allocation must satisfy $x_A \le 11,\ x_B \le 7,\ x_A + x_B = 15$. The Shapley value lands inside such limits in convex cost games.

2.3 Bankruptcy Problems

The bankruptcy problem (N, A, c)

A debtor's assets total A, but the sum of creditor claims $C = \sum_i c_i$ exceeds A (that's what "bankrupt" means). A **solution** allocates x_i to each creditor with $0 \le x_i \le c_i$ and $\sum_i x_i = A$. We model it as a cooperative game where each creditor's claim is c_i and $v(N) = A$.

Two competing rules (Example 7: $A = 100$, $c_1 = 50$, $c_2 = 100$)

Proportional rule: each creditor gets the *same fraction* A/C of her claim. Here $C = 150$, $A/C = \frac{2}{3}$:

$$x_1 = \frac{2}{3} \cdot 50 = \frac{100}{3} \approx 33.3, \qquad x_2 = \frac{2}{3} \cdot 100 = \frac{200}{3} \approx 66.7.$$

Contested-garment (Talmud) rule: each creditor first concedes to the other whatever she does *not* claim; the *contested* remainder is split equally.

$$\text{conceded to } 1 = A - c_2 = 100 - 100 = 0, \quad \text{conceded to } 2 = A - c_1 = 100 - 50 = 50,$$

$$\text{contested} = A - (0 + 50) = 50, \text{ split equally} \Rightarrow x_1 = 0 + 25 = 25, \ x_2 = 50 + 25 = 75.$$

The contested-garment principle (the Talmud)

"Two hold a garment; *both* claim it all $\Rightarrow$ each gets half." "One claims it *all*, the other claims *half* $\Rightarrow$ $\frac{3}{4}$ to the first, $\frac{1}{4}$ to the second." The rule: you are awarded everything the other side *concedes* (doesn't claim), plus half of what is jointly contested. Remarkably, applied pairwise-consistently this rule coincides with the **Shapley value** (equivalently the *nucleolus*) of the bankruptcy game — the ancient Talmudic answer is the modern game-theoretic one.

The two Talmudic garment cases, derived

A garment worth 1. **Case A — both claim all:** each concedes 0 to the other; contested $= 1$; split equally $\Rightarrow (\frac{1}{2}, \frac{1}{2})$. **Case B — one claims all, the other claims half:** the half-claimant concedes $\frac{1}{2}$ to the full-claimant (the part she doesn't claim); the full-claimant concedes 0. Contested $= 1 - \frac{1}{2} = \frac{1}{2}$, split equally ($\frac{1}{4}$ each). So the full-claimant gets $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ and the half-claimant gets $0 + \frac{1}{4} = \frac{1}{4} \Rightarrow (\frac{3}{4}, \frac{1}{4})$. This $\frac{3}{4}/\frac{1}{4}$ rule is the signature of the contested-garment principle.

Sharing infrastructure between communities (core limits)

Communities A and B can each build a facility alone for 11 and 7; together for 15. An *equal* split (7.5 each) is rejected by B (it can go alone for 7). A core cost allocation must satisfy $x_A + x_B = 15$, $x_A \leq 11$, $x_B \leq 7 \Rightarrow x_A \in [8, 11]$, $x_B \in [4, 7]$. The Shapley value splits the joint saving evenly: standalone total $11 + 7 = 18$, joint 15, saving 3, so each saves $1.5 \Rightarrow x_A = 9.5$, $x_B = 5.5$ — inside the core. With three communities, add the analogous standalone constraints for every coalition.

★ Exam Tip

Know *both* rules and when each is asked. **Proportional** = same fraction of every claim ($x_i = \frac{A}{C} c_i$). **Contested-garment** = concede-then-split-contested (gives the small creditor relatively *more* when assets are scarce, the large creditor relatively more when assets are plentiful). For two equal claimants both rules agree (split equally).

Common Mistake

Do not confuse the contested-garment award with "equal split" — they coincide only in special cases. And remember the feasibility bounds $0 \leq x_i \leq c_i$: a creditor never receives more than she claimed, and the awards must sum to the available assets A, not to the total claims C.

2.4 More Worked Examples and Methods

The partnership, computed in full (all six orders)

Game $v(1) = 30, v(2) = 12, v(3) = 6,\ v(12) = v(13) = 36,\ v(23) = 30,\ v(123) = 90$. Pay each entrant her marginal contribution against those already present:

Order	MC_1	MC_2	MC_3
123	30	36−30=6	90−36=54
132	30	90−36=54	36−30=6
213	36−12=24	12	90−36=54
231	90−30=60	12	30−12=18
312	36−6=30	90−36=54	6
321	90−30=60	30−6=24	6
Sum	234	162	144
÷6	**39**	**27**	**24**

$\boxed{\phi = (39, 27, 24)}$, summing to $90 = v(N)$. The "dissolving a partnership" procedure (claims taken in increasing coalition size, each coalition's worth split equally among its members and netted out of larger coalitions) reaches this *same* vector — and, by Result 1, does so for *any* claim order. The order table is the safest way to be sure.

The airport runway game (classic cost-sharing)

Three airlines share one runway. Airline 1 needs a short runway costing 10; airline 2 needs a medium one costing 20; airline 3 needs a long one costing 30 (the long runway serves everyone). Build it as nested arcs: the first 10 of length is used by *all three*, the next 10 (to reach 20) by airlines 2 and 3, the final 10 (to reach 30) by airline 3 alone.

$$\phi_1 = \tfrac{10}{3} = 3.33, \quad \phi_2 = \tfrac{10}{3} + \tfrac{10}{2} = 8.33, \quad \phi_3 = \tfrac{10}{3} + \tfrac{10}{2} + 10 = 18.33.$$

Sum $= 30$ ✓. "Each segment split equally among those who use it" is the Shapley cost rule — the cheaper-needs airline subsidises none of the long-runway cost.

The Talmud estate table (Aumann–Maschler)

Two creditors claim 100 and 200; apply the contested-garment rule as the estate A grows:

Estate A	Creditor 1 (claim 100)	Creditor 2 (claim 200)	Behaviour
100	50	50	*equal* split (both fully contest)
200	50	150	creditor 1 capped near claim/2
300	100	200	*proportional* once assets are ample

Small estate ⇒ near-*equal* division (egalitarian); large estate ⇒ near-*proportional*. The contested-garment rule interpolates between the two — which is exactly why it "felt fair" to the Talmudic sages and matches the modern nucleolus.

Fairness criteria and cake-cutting

For dividing a continuous "cake" among n people:

- **Proportional:** everyone believes they got $\geq \frac{1}{n}$ of the value.
- **Envy-free:** nobody prefers another's piece to their own (stronger; implies proportional).

- **Efficient (Pareto):** no other division makes someone better off without hurting another.

Divide-and-choose (one cuts, the other picks) is envy-free for $n = 2$: the cutter makes two pieces she values equally; the chooser takes her favourite. For general n, the *last-diminisher* procedure guarantees proportionality.

Three-creditor contested garment

With three or more creditors, apply the contested-garment principle *pairwise-consistently* (the unique allocation that is CG-consistent for every pair). This coincides with the *nucleolus* and the Shapley value of the bankruptcy game — the same answer the Talmud prescribes.

End of Week 10

Quick Revision — Week 10

Dissolving a partnership: the claims procedure ⇒ Shapley value (order-independent). **Cost games:** model cost as $v(S)$; for tree/network costs the Shapley value = split each arc equally among its users; respect standalone (core) limits. **Bankruptcy** (N, A, c) with $C = \sum c_i > A$: *proportional* rule $x_i = \frac{A}{C} c_i$; *contested-garment* rule (concede then split the contested half), which equals the Shapley value/nucleolus.

Formula Sheet — Week 10

$$\text{Proportional: } x_i = \frac{A}{C} c_i; \qquad \text{Tree cost share: } \phi_i = \sum_{\text{arcs } i \text{ uses}} \frac{\text{arc cost}}{\#\text{users of arc}}.$$

Contested garment (2 claimants): $x_i = (A - c_j)^+ + \frac{1}{2}\big[A - (A - c_i)^+ - (A - c_j)^+\big]$.

Practice questions.

1. Bankruptcy $A = 100$, claims $(c_1, c_2) = (50, 100)$: confirm proportional $(\frac{100}{3}, \frac{200}{3})$ and contested-garment $(25, 75)$.
2. If assets rise to $A = 200$ with the same claims, recompute both rules. Which creditor gains relatively?
3. TV-cable game: verify the equal-split-per-arc allocation $(11, 14, 35)$ sums to the total cost 60.
4. Explain why the partnership claims procedure gives the same answer regardless of claim order.
5. State the standalone constraints that any acceptable cost split among three communities must satisfy.

CHAPTER 3

Week 11 — Auctions

Why auctions?

An auction is the simplest market with *hidden information*: a seller faces buyers whose values she cannot see. The auction *format* determines how those values get revealed and who pays what. Auctions are everywhere — eBay, Google ads, spectrum, IPL, art — and they are the cleanest setting to apply the Bayesian-game tools of Quiz 2.

3.1 The Four Classic Formats

The formats

A single item; each bidder i has a private **value** v_i (the most she'd pay).

- **English (ascending):** price rises; bidders drop out; last one standing wins at the price where the second-last dropped.
- **Dutch (descending):** price falls from a high start; the first bidder to accept wins at that price.
- **First-price sealed bid:** simultaneous sealed bids; highest bid wins and *pays its own bid.*
- **Second-price sealed bid (Vickrey):** highest bid wins but *pays the second-highest bid.*

Two strategic equivalences

Dutch $\equiv$ First-price: in both, you must commit to a single number not knowing others' — the clock's stopping point *is* your sealed bid. **English $\equiv$ Second-price:** in both, you stay in until the price reaches your value, so the winner effectively pays (just above) the second-highest value. These pairings let us analyse just two cases.

Private vs. common values

Independent private values: each bidder knows her own value, which is unaffected by others' (a painting you'll keep). **Common value:** the item is worth the *same* (unknown) amount to all (an oilfield); each bidder has only a noisy estimate — the setting of the *winner's curse.*

3.2 Second-Price (Vickrey): Truthful Bidding Is Dominant

Bidding your true value weakly dominates

Fix bidder i with value v; let h = highest *rival* bid. Your bid b affects only *whether* you win, never *what* you pay (you always pay h if you win). Compare $b = v$ to any deviation:

- **Raising** $b > v$: changes the outcome only if $v < h < b$ — you now win but pay $h > v$, a loss. Never helps.
- **Lowering** $b < v$: changes the outcome only if $b < h < v$ — you now lose a sale you'd have made at a profit. Never helps.

So $b_i = v_i$ **is a (weakly) dominant strategy** — regardless of what anyone else does. No guessing, no shading. (English auctions inherit this: stay in until the price hits your value.)

Equilibrium Insight

Vickrey's beauty: it is *strategy-proof*, so it elicits true values and the item goes to the bidder who values it most (*efficient*). This is exactly the principle the sponsored-search VCG mechanism (Week 12) generalises.

3.3 First-Price: Shade Your Bid

Simple Intuition

Bidding your value guarantees zero profit (you pay exactly what it's worth). So you *shade* — bid below value — trading a lower chance of winning for a positive margin if you do. How much to shade depends on how many rivals you face.

Equilibrium with n bidders, values uniform on $[0, 1]$

Guess a symmetric strategy $b(v) = \alpha v$. A bidder with value v bidding b wins iff all $n - 1$ rivals have values below b/α, with probability $(b/\alpha)^{n-1}$. Expected payoff $(v - b)(b/\alpha)^{n-1}$; maximising over b gives

$$\boxed{b(v) = \frac{n-1}{n}\, v}.$$

You bid the fraction $\frac{n-1}{n}$ of your value: shade a lot with few rivals ($v/2$ for $n = 2$), barely at all with many ($\to v$ as $n \to \infty$). More competition $\Rightarrow$ less shading.

3.4 Order Statistics and Revenue Equivalence

The one fact you need: expected order statistics

If n values are drawn independently and uniformly on $[0, 1]$ and sorted, the expected value of the k-th *smallest* is

$$\mathbb{E}[\,k\text{-th smallest}\,] = \frac{k}{n+1}.$$

Hence $\mathbb{E}[\text{highest}] = \dfrac{n}{n+1}$ and $\mathbb{E}[\text{second-highest}] = \dfrac{n-1}{n+1}$.

Both auctions give the seller the same revenue

Second-price: the winner pays the second-highest *value*, so expected revenue $= \mathbb{E}[\text{2nd-highest}] = \dfrac{n-1}{n+1}$. **First-price:** each bidder bids $\frac{n-1}{n}v$, so the seller collects $\frac{n-1}{n} \times$ (highest value); expected revenue $= \frac{n-1}{n} \cdot \mathbb{E}[\text{highest}] = \frac{n-1}{n} \cdot \frac{n}{n+1} = \dfrac{n-1}{n+1}$.

$$\boxed{\text{Same expected revenue } = \frac{n-1}{n+1}.}$$

Revenue Equivalence Theorem

Under independent private values, risk-neutral bidders, and symmetric value distributions, *all four standard auctions yield the seller the same expected revenue.* The shading in first-price exactly offsets the fact that the winner pays her own (higher) bid. Format choice is then about *robustness, simplicity, and collusion-resistance*, not expected revenue.

★ Reserve prices

A seller can do *better* than revenue equivalence by setting a **reserve price** r (sell only if the top bid $\geq r$). The reserve risks losing a sale but extracts more when values are high; the optimal reserve is strictly positive and *independent of the number of bidders.* "Set a reserve" is the standard answer to "how can the seller raise revenue?"

3.5 The Winner's Curse (Common Values)

Simple Intuition

In a common-value auction (everyone's value is the same unknown V, each with a noisy estimate), *winning is bad news*: you win precisely when your estimate was the most *optimistic*, i.e. likely too high. Naively bidding your estimate leads to systematic overpayment — the **winner's curse**.

Equilibrium Insight

The remedy is to **shade your bid down further** than in a private-value auction, conditioning on the sobering thought "if I win, my estimate was probably the highest, so the truth is likely lower." This is the same "condition on winning" logic as the adverse-selection problem in Book 1.

3.6 The All-Pay Auction and a Format Comparison

All-pay auction

Every bidder submits a bid; the highest bid wins the item; **but everyone pays their bid regardless of winning.** Models lobbying, R&D races, and contests where effort is sunk whether or not you win.

Equilibrium Insight

Because losing bids are forfeited, bidders *shade aggressively* (bid low) — you don't want to pour money in and lose. With n bidders, values uniform on $[0,1]$, the symmetric equilibrium bid is $\beta(v) = \frac{n-1}{n}v^n$, and — remarkably — the seller's expected revenue is again $\frac{n-1}{n+1}$. **Revenue equivalence holds even here:** despite radically different rules, expected revenue is unchanged.

Format	Who pays	Equilibrium bid	Truthful?
Second-price / English	winner pays 2nd bid	bid your value v	Yes (dominant)
First-price / Dutch	winner pays own bid	$\frac{n-1}{n}v$ (shade)	No
All-pay	*everyone* pays bid	$\frac{n-1}{n}v^n$	No

When revenue equivalence breaks

The theorem needs *risk-neutral* bidders, *independent private* values, and *symmetry*. Relax these and formats diverge: with **risk-averse** bidders the first-price auction raises *more* (bidders bid higher to reduce the risk of losing); with **common values** the winner's curse depresses bids; with **asymmetric** bidders efficiency and revenue rankings can flip. "Revenue equivalence, *under its assumptions*" is the precise exam phrasing.

3.7 Worked Practice Problems

Chandan's bakery cake

Five bidders value the cake at: Amar 1000, Zaheer 200, and three others 50 each.

- **English auction:** bidders drop out as the price climbs; everyone but Amar quits by ≈ 200 (Zaheer's value). **Amar wins and pays a bit more than 200.**
- **Second-price (no one knows others' values):** truthful bidding is dominant, so Amar bids 1000, Zaheer 200, others 50. **Amar wins and pays the second-highest bid $= 200$.**

English and second-price give the *same* outcome (winner Amar, price $\approx$ 200) — the equivalence in action.

★ Koyal's tractor (reserve-price problem)

Two bidders, values 6,00,000 (professional) or 4,50,000 (recreational). Under a second-price/Vickrey sale each bids truthfully; the seller receives the *second-highest* value (e.g. two professionals $\Rightarrow$ pays 6,00,000; one of each $\Rightarrow$ pays 4,50,000). To raise expected revenue, set a *reserve* between the two possible values so a lone high-value buyer still pays near her value. The method: enumerate the value combinations, weight by probabilities, and compare expected revenue with and without the reserve.

3.8 More Worked Examples

First-price equilibrium, derived cleanly (n bidders, uniform)

Conjecture a symmetric increasing bid $b(v)$. If your value is v and you bid as if your value were z (i.e. bid $b(z)$), you win when all $n-1$ rivals have values below z — probability z^{n-1} — and pay $b(z)$. Expected payoff

$$\pi(z) = (v - b(z))\, z^{n-1}.$$

Optimality requires $z = v$ at the maximum: $\frac{d\pi}{dz}\big|_{z=v} = 0$ gives the differential equation $b'(v)v^{n-1} + b(v)(n-1)v^{n-2} = v \cdot (n-1)v^{n-2}$, i.e. $\frac{d}{dv}[b(v)v^{n-1}] = (n-1)v^{n-1}$. Integrating from 0:

$$b(v)v^{n-1} = (n-1)\int_0^v t^{n-1}dt = \tfrac{n-1}{n}v^n \;\Rightarrow\; \boxed{b(v) = \tfrac{n-1}{n}v}.$$

This confirms the guess-and-verify result and shows *why* the shading factor is exactly $\frac{n-1}{n}$.

Optimal reserve price (single bidder, value uniform $[0,1]$)

A seller (cost 0) faces one buyer with value $v \sim U[0,1]$. With *no* reserve the buyer pays ≈ 0. Set a reserve r: a sale happens iff $v \geq r$ (probability $1-r$) at price r. Expected revenue

$$R(r) = r(1-r), \qquad R'(r) = 1 - 2r = 0 \Rightarrow \boxed{r^* = \tfrac{1}{2}}, \;\; R = \tfrac{1}{4}.$$

A reserve of $\frac{1}{2}$ turns near-zero revenue into $\frac{1}{4}$. The optimal reserve is *positive* and (with more bidders) *independent of* n — it trades the risk of no sale against a higher price when values are high.

All-pay auction revenue equals the others ($n = 2$, uniform)

Symmetric equilibrium bid $\beta(v) = \frac{n-1}{n}v^n = \frac{1}{2}v^2$. Each bidder's *expected payment* (they pay regardless of winning) is

$$\int_0^1 \tfrac{1}{2}v^2\,dv = \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{1}{6}, \qquad \text{so seller revenue} = 2 \times \tfrac{1}{6} = \tfrac{1}{3} = \tfrac{n-1}{n+1}.$$

Same $\frac{1}{3}$ as the first- and second-price auctions — **revenue equivalence** confirmed by direct computation, despite everyone paying.

Winner's curse with numbers

An oil tract is worth a *common* V, equally likely \$10M or \$30M (mean \$20M). Each of 10 firms gets a noisy private estimate. Bidding your *unconditional* mean (\$20M) ignores a key fact: *you only win when your estimate is the highest*, which happens disproportionately when V is truly low *and* your noise was high. Conditioning on "I won" lowers the expected value below \$20M, so a naive bidder *overpays and loses money on average*. **Fix:** shade your bid below your estimate by the expected over-optimism of the winner — the more rivals, the larger the shade.

End of Week 11

Quick Revision — Week 11

Four formats: English/ascending, Dutch/descending, first-price, second-price. **Equivalences:** Dutch≡first-price, English≡second-price. **Second-price/Vickrey:** bid truthfully (dominant) ⇒ efficient. **First-price (uniform $[0,1]$):** bid $\frac{n-1}{n}v$ (shade; less with more rivals). **Revenue equivalence:** all four give expected revenue $\frac{n-1}{n+1}$; a *reserve price* beats this. **Winner's curse:** common values ⇒ shade down more.

Formula Sheet — Week 11

$b(v) = \frac{n-1}{n}v$; $\mathbb{E}[k\text{-th smallest of } n \text{ uniform}] = \frac{k}{n+1}$; $\mathbb{E}[\text{2nd-highest}] = \frac{n-1}{n+1} = \text{expected revenue.}$

★ Algorithm sheet

"How to bid?" second-price ⇒ value; first-price ⇒ $\frac{n-1}{n}v$. **"Who wins / what price?"** highest value wins; price = second-highest (English/second) or own shaded bid (first/-Dutch). **"Expected revenue?"** $\frac{n-1}{n+1}$ (uniform). **"Raise revenue?"** reserve price. **"Common value?"** beware winner's curse, shade more.

Practice questions.

1. Three bidders, uniform $[0, 1]$: equilibrium first-price bid? Expected revenue? *[$\frac{2}{3}v$; $\frac{2}{4} = \frac{1}{2}$.]*
2. Prove second-price truthful bidding is dominant by the two-case argument.
3. Bakery cake: who wins and what is paid under a *first-price* auction if values are common knowledge?
4. Why does revenue equivalence fail if bidders are risk-averse?
5. Explain the winner's curse and the correct bidding adjustment.

CHAPTER 4

Week 12 — Network Effects and Sponsored Search

The information economy

Choosing potatoes vs. onions, you only weigh your own taste. Choosing WhatsApp vs. Telegram, you care intensely about *what everyone else chooses* — a product is more valuable the more people use it. That is a **network effect**, and it drives tipping, lock-in, and winner-take-all markets. The chapter ends with **sponsored search**, where the same matching/auction ideas price billions of ad slots.

4.1 Network Effects: Direct and Indirect

Network effect

A product exhibits a **network effect** if each user's payoff *increases in the number of other users* of that product (or of compatible products). Two channels:

- **Direct** (network/communication markets): more users ⇒ more people to interact with ⇒ more reason to join. *Phones, fax, email, messaging, a shared language.*
- **Indirect** (system markets, "hardware + software"): more users ⇒ more developers write complements ⇒ more reason to buy. *Game consoles, smartphones, operating systems.*

Switching costs and lock-in

Switching costs are the costs of moving to a functionally identical rival (re-learning, lost data, lost compatibility). They create **lock-in** and hand *market power* to incumbents. Two further features dominate these markets: **expectations** (what users *think* others will do is self-fulfilling) and **history** (past adoptions and market shares are valuable assets — *path dependence*).

4.2 Modelling Demand with Network Effects

Simple Intuition

Each consumer's willingness to pay rises with the expected network size. We need an equilibrium in which *expectations are fulfilled* — the number who actually buy equals the number everyone expected to buy.

Reservation prices and fulfilled expectations

Index consumers $x \in [0,1]$ in *decreasing* order of reservation price $r(x)$ (so lower x = keener buyer). A consumer's value has a *standalone* part plus a *network* part proportional to the expected fraction n_e who buy. With a linear specification, the price the marginal consumer n will pay, given expected size n_e, is

$$p(n, n_e) = a + v\, n_e\, (1-n),$$

and at a **fulfilled-expectations** (rational-expectations) equilibrium $n = n_e$:

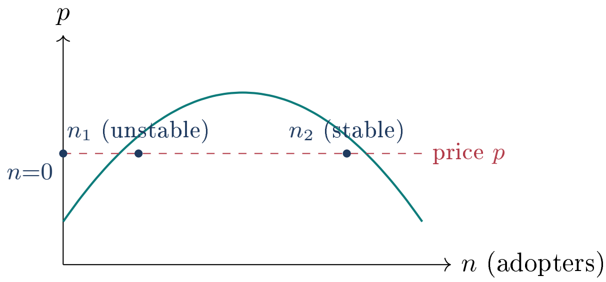
$$\boxed{p(n, n) = a + v\, n\, (1-n)}.$$

Here a is the standalone value, v scales the network benefit, and $(1-n)$ reflects that later adopters value the good less.

The hump, and why markets "tip"

The fulfilled-expectations demand $p(n,n) = a + vn(1-n)$ is a **downward-then-upward then downward** (hump-shaped) curve, not a simple downward slope. Consequently, *at a single price p there can be three equilibria*:

- $n = 0$: **nobody buys** — and it's self-fulfilling (no network, no reason to join). **Stable.**
- $n_1(p)$: a **small** adoption — **unstable** (the *critical mass*/tipping point): drop slightly below and the market collapses to 0; rise slightly above and it grows.
- $n_2(p)$: a **large** adoption — **stable** (the thriving network).



Coordination, lock-in, standards wars

Because expectations are self-fulfilling, a superior new product can fail purely from a **coordination failure** (everyone waits for everyone else). In a standards war (format A vs. B), the market typically **locks in to a monopoly** of one standard, and *which* one can be unpredictable (path-dependent on early, possibly accidental, leads — think QWERTY, VHS vs. Betamax). Implications for firms: *manage expectations*, subsidise early adopters to reach critical mass, and value installed base.

★ Exam Tip

Exam hooks: identify *direct* vs. *indirect* effects; explain *critical mass / tipping*; read the three equilibria off the hump and label their stability; and argue why network markets tend to *winner-take-all* with lock-in. "Expectations are self-fulfilling" and "history matters" are the two phrases graders look for.

Benchmark: efficiency *without* network effects

Strip out the network term ($v = 0$). Consumers buy whenever $r(x) \geq p$, so at the market price p^* exactly the keenest fraction x^* buy. This equilibrium is **socially optimal**: social welfare is the area between the reservation-price curve $r(x)$ and price, and it is maximised precisely by serving consumers 0 through x^* (each added buyer beyond x^* values the good *less* than its price). Competition drives any above-p^* price to zero profit. *With* network effects this clean efficiency breaks: the wrong (low) equilibrium can persist, so coordination/expectations management has real welfare value.

	Direct network effect	Indirect network effect
Source of value	Interacting with other users	Complementary products (software)
Market type	Network / communication	System (hardware + software)
Mechanism	more users $\Rightarrow$ more contacts	more users $\Rightarrow$ more complements
Examples	phone, email, messaging, language	consoles, smartphones, OS

4.3 Sponsored Search as a Matching Market

Simple Intuition

A search engine sells several ad **slots** per query; higher slots get more clicks. Advertisers value a click differently. This is a *matching market* (assign slots to advertisers) combined with an *auction* (set prices when values are private).

The setup

Slots $a, b, c, \ldots$ have **clickthrough rates** (clicks/hour) decreasing down the page; crucially the clickthrough rate depends *only on the slot*, not on which ad sits there. Advertiser i has a private **value per click** v_i, so her value for a slot $= v_i \times$ (that slot's clickthrough rate). Pricing is **cost-per-click (CPC)**: you pay only when clicked.

Market-clearing prices

Set per-slot prices so that, when each advertiser picks the slot maximising (value − price), the resulting "preferred-slot" graph has a **perfect matching** — every advertiser gets a distinct slot she most prefers. Such prices *clear the market*. (A constructive algorithm raises prices on over-demanded sets of slots until a perfect matching appears.)

The VCG mechanism (truthful pricing)

When values are *private*, use **Vickrey–Clarke–Groves**: (1) collect bids; (2) choose the assignment that **maximises total value**; (3) charge each advertiser the **externality** she imposes — the total value the *others* lose by her presence. This makes **truthful bidding a dominant strategy** (the single-slot case is exactly the second-price auction) and yields an *efficient* allocation.

A three-slot VCG computation (illustrative)

Slots a, b, c with clickthrough rates $10, 5, 2$. Advertisers x, y, z with per-click values $3, 2, 1$. **Efficient assignment** (highest value to highest clicks): $x \to a,\ y \to b,\ z \to c$, total value

$3(10) + 2(5) + 1(2) = 42$. **VCG charge = harm to others:**

$$x \text{ pays } \Big[\underbrace{2(10) + 1(5)}_{\text{others best without } x} \Big] - \Big[\underbrace{2(5) + 1(2)}_{\text{others with } x} \Big] = 25 - 12 = 13,$$

y pays $[3(10)+1(5)]-[3(10)+1(2)] = 35-32 = 3$, z pays $[3(10)+2(5)]-[3(10)+2(5)] = 40-40 = 0$.

So x, y, z pay $13, 3, 0$ total (per-click 1.3, 0.6, 0). Each pays *less than her own value* and exactly the cost she imposes on the rest — the hallmark of VCG. *(Numbers illustrative; the method is the point.)*

GSP vs. VCG

Real engines historically used the **Generalized Second-Price (GSP)** auction (each slot pays the next bid down), which is *not* truthful in general, whereas **VCG** is. Both reduce to the ordinary second-price auction when there is a single slot. The exam point: VCG = assign efficiently + charge the externality = truthful.

4.4 More Worked Examples

Solving for the three network equilibria (with stability)

Take $a = 0$, $\nu = \frac{1}{3}$ so the fulfilled-expectations demand (Form A) is $p = n(1-n)$. Suppose the firm sets price $p = 0.16$. Equilibria solve $n(1-n) = 0.16$:

$$n^2 - n + 0.16 = 0 \;\Rightarrow\; n = \frac{1 \pm \sqrt{1 - 0.64}}{2} = \frac{1 \pm 0.6}{2} \;\Rightarrow\; n = 0.2 \text{ or } 0.8,$$

plus the trivial $n = 0$. **Three equilibria:** $0,\ 0.2,\ 0.8$. **Stability:** just above $n = 0.2$ the value consumers place on joining exceeds the price, so more join and the network climbs to 0.8; just below 0.2 it collapses to 0. Hence 0 and 0.8 are *stable*, and 0.2 is the *unstable critical mass* (tipping point). To launch the product the firm must push adoption *past* 0.2 — via subsidies, free trials, or expectations management — after which growth to 0.8 is self-sustaining.

Monopoly pricing of a network good

A monopolist (zero cost) sets p; buyers coordinate on the high stable equilibrium $n_2(p)$. Revenue $= p \cdot n_2(p)$. With $p = n(1-n)$ the firm chooses the operating point n to maximise $p \cdot n = n^2(1-n)$: $\frac{d}{dn}[n^2 - n^3] = 2n - 3n^2 = 0 \Rightarrow n = \frac{2}{3}$, giving $p = \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9}$ and revenue $\frac{2}{9} \cdot \frac{2}{3} = \frac{4}{27}$. **The monopolist deliberately serves *fewer* than the maximum** to keep the price up — but must still clear the critical mass to avoid collapse.

Sponsored search: full VCG computation

Slots a, b with clickthrough $10, 5$; advertisers x, y, z value a click at $3, 2, 1$. **Step 1 — efficient assignment:** highest value to most clicks: $x \to a,\ y \to b,\ z$ out. Total social value $= 3(10) + 2(5) = \mathbf{40}$. **Step 2 — VCG price = externality on others:**

$$\text{slot } a\ (x):\ \underbrace{\big[y \to a,\ z \to b\big]}_{2(10)+1(5)=25} - \underbrace{\big[y \to b\big]}_{2(5)=10} = \mathbf{15},$$

$$\text{slot } b\ (y):\ \underbrace{[x \to a,\ z \to b]}_{3(10)+1(5)=35} - \underbrace{[x \to a]}_{3(10)=30} = \mathbf{5}.$$

So x pays 15 (i.e. 1.5/click), y pays 5 (1/click). Each pays *below* her own value and exactly the harm she imposes — truthful bidding is optimal.

GSP vs. VCG — why GSP is not truthful

Generalized Second-Price (GSP): each winner pays (per click) the *bid of the advertiser just below*. With truthful bids $3, 2, 1$: x (slot a) pays 2/click, y (slot b) pays 1/click. Compare the **VCG** per-click prices 1.5 and 1. They differ (x pays 2 under GSP vs. 1.5 under VCG), so under GSP advertisers have an incentive to *shade* their bids — GSP is *not* truthful, while VCG is. Both coincide with the ordinary second-price auction when there is a single slot.

Market-clearing prices by the ascending procedure

To price slots when values are private: start all slot prices at 0; each advertiser points to the slot maximising (value−price). If the "preferred-slot" graph has a perfect matching, those prices clear the market. If a set of slots is *over-demanded* (more advertisers point to it than it contains), raise the prices on that set by one unit and repeat. The procedure terminates at market-clearing prices that support an efficient assignment — the discrete analogue of supply-equals-demand, and the foundation under the VCG result above.

End of Week 12

Quick Revision — Week 12

Network effect: payoff rises with #users. *Direct* (communication) vs. *indirect* (hardware+software). **Switching costs** ⇒ **lock-in**; expectations self-fulfilling; history (path dependence) matters. **Fulfilled-expectations demand** $p(n, n) = a + vn(1 - n)$ is hump-shaped ⇒ three equilibria: 0 (stable), n_1 (unstable critical mass), n_2 (stable) ⇒ *tipping*. **Sponsored search:** slots × clickthrough; **VCG** assigns to maximise total value and charges each advertiser the externality ⇒ truthful & efficient; GSP is not truthful.

Formula Sheet — Week 12

$$p(n,n) = a + vn(1-n); \qquad \text{VCG charge}_i = \Big[\text{others' best value without } i\Big] - \Big[\text{others' value with } i\Big].$$

★ Algorithm sheet

Network demand: write $p(n, n) = a + vn(1 - n)$; set $= p$ to find the (up to three) equilibria; label stability by the slope. **VCG:** (1) assign value-rank to click-rank; (2) for each advertiser, recompute the others' best assignment *without* her; (3) charge the difference.

Practice questions.

1. Classify direct vs. indirect: fax machine, app store, credit-card network, spoken language.
2. With $a = 0, v = 1$, solve $p = n(1 - n)$ for $p = 0.16$; identify the critical mass and the stable network.
3. Recompute the three-slot VCG prices if z's per-click value rises to 4 (re-rank first!).

4. Explain why a better product can lose a standards war.
5. Show the single-slot VCG charge equals the second-highest bid.

Part II

Complete PYQ Solutions

How Part II Works

The End-Term is *cumulative*. Across the eight papers (2024 T2/T3 and 2025 T2/T3, each in forenoon/afternoon sittings) the *same question families* recur with changed numbers. Every family is solved below with the official answer keys reproduced and independently re-derived; cumulative Quiz-1/Quiz-2 mechanics are cross-referenced to Books 1–2.

★ The fourteen End-Term families

New (Weeks 9–12): (1) network-effects demand, (2) Shapley value, (3) Shapley–Shubik power index, (4) cable-TV cost-sharing, (5) bankruptcy/contested-garment, (6) first-price auction, (7) sponsored-search VCG, (8) order statistics. **Cumulative:** (9) Bayes' rule, (10) the campaign repeated game, (11) roommate/effort best-response, (12) weak-dominance & a killed NE, (13) Gale–Shapley matching, (14) a symmetric mixed-entry / declining-industry game. A few cross-listed *design-thinking* and *credit-analytics* items also appear (scope note at the end).

CHAPTER 5

Family 1 — Network-Effects Demand Curve

Frequently Asked PYQ Concept

A consumer of type $\theta \sim U[0,1]$ joins a network good. Two utility forms appear; derive the indifferent consumer, the mass of buyers, the maximum price, and the fulfilled-expectations demand curve.

5.1 Form A (additive): $U(\theta) = a + 3\theta\nu n^e$

Indifference $U(\hat{\theta}) = p$: $\hat{\theta} = \dfrac{p-a}{3\nu n^e}$. Buyers are $\theta \geq \hat{\theta}$ so mass $n = 1 - \hat{\theta}$. Rational expectations $n^e = n$:

$$\boxed{p = a + 3\nu\, n(1-n)}, \qquad p_{\max} = a + \tfrac{3\nu}{4} \text{ at } n = \tfrac{1}{2}.$$

5.2 Form B (multiplicative): $U(\theta) = \theta\,(a + \nu n^e)$

Indifference: $\hat{\theta} = \dfrac{p}{a + \nu n^e}$. **Maximum price to have any buyers** ($\theta = 1$): $p_{\max} = a + \nu n^e$. Mass $n = 1 - \hat{\theta}$. Rational expectations:

$$\boxed{p = (a + \nu n)(1-n)}.$$

Verified answers

Form A: $\hat{\theta} = \frac{p-a}{3\nu n^e}$, demand $p = a + 3\nu n(1-n)$. Form B: $\hat{\theta} = \frac{p}{a+\nu n^e}$ (keyed), $p_{\max} = a + \nu n^e$ (keyed), mass $= 1 - \hat{\theta}$ (keyed), demand $p = (a+\nu n)(1-n)$. Both demand curves are *hump-shaped* $\Rightarrow$ three equilibria (0, unstable critical mass, stable network): **tipping**.

Quiz Trap

Impose $n^e = n$ *after* writing the indifference condition. Keep the $(1-n)$ factor — it produces the hump and the tipping behaviour. Read whether utility is *additive* $(a + 3\theta\nu n^e)$ or *multiplicative* $(\theta(a + \nu n^e))$; the $\hat{\theta}$ formula differs.

CHAPTER 6

Family 2 — Shapley Value

Frequently Asked PYQ Concept

"Calculate the Shapley value of the following game" (sometimes "using the procedure for dissolving a partnership"). Average each player's marginal contribution over all $n!$ entry orders; for $n = 3$ use the one-line formula.

$$\phi_i^{(n=3)} = \frac{v(i)}{3} + \frac{(v(ij) - v(j)) + (v(ik) - v(k))}{6} + \frac{v(N) - v(jk)}{3}.$$

Verified instances

Partnership $v(1) = 30, v(2) = 12, v(3) = 6, v(12) = v(13) = 36, v(23) = 30, v(123) = 90$ $\Rightarrow \phi = (39, 27, 24)$. A second paper's partnership keys to $\phi \approx (16.7, 36.7, 6.7)$ (sum 60). Example 18 (Week 9) $\Rightarrow (22, 40, 58)$. Always confirm $\sum_i \phi_i = v(N)$, and use symmetry/null shortcuts.

Verified answers

Garage–petrol–restaurant: $(39, 27, 24)$. The dissolving-partnership claims procedure gives *exactly* the Shapley value, independent of claim order.

CHAPTER 7

Family 3 — Shapley–Shubik Power Index $[61; 40, 40, 30, 10]$

Quota 61; weights $(40, 40, 30, 10)$. Winning $(v = 1)$: every pair among $\{1, 2, 3\}$, all triples, the grand coalition; everything else 0.

Verified answers

Coalition function: $v(\text{singletons}) = 0$; $v(12) = v(13) = v(23) = 1$, $v(14) = v(24) = v(34) = 0$; *all* triples and grand $= 1$. **Symmetric players:** $\{1, 2, 3\}$. **Null player:** $\{4\}$ (weight 10 is never pivotal). **Power index** $\phi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0)$.

Quiz Trap

Power $\neq$ weight: the 10-weight member has *zero* power. A null player always scores 0; identify null/symmetric players before computing.

CHAPTER 8

Family 4 — Cable-TV Cost-Sharing

Frequently Asked PYQ Concept

A cable station connects customers along a cost tree; share each arc's cost *equally among the customers who use it* (this equals the Shapley value of the cost game).

Verified answers (per paper)

Cost shares (player 1, 2, 3): 2024 T2 $\to (15, 15, 20)$; 2024 T3 $\to (6, 28, 30)$; 2025 T2 $\to (8, 30, 32)$. A six-customer variant keys to $(7, 12, 12, 17, 18, 19)$. In every case the shares sum to the total network cost.

★ Exam Tip

Method: for each arc, divide its cost by the number of downstream customers who use it; each customer pays the sum of her per-arc shares. Shared trunk arcs are split among *all* users; private "last-mile" arcs are paid fully by that one customer.

CHAPTER 9

Family 5 — Bankruptcy / Contested Garment

Frequently Asked PYQ Concept

Estate A below total claims; divide by the *contested-garment* (Talmud) principle: each gets what the other concedes, plus half the contested remainder.

Estate 500, claims $(400, 300)$

Conceded to 1: $A - c_2 = 500 - 300 = 200$. Conceded to 2: $A - c_1 = 500 - 400 = 100$. Contested $= 500 - 200 - 100 = 200$, split equally (100 each). So $x_1 = 200 + 100 = \boxed{300}$, $x_2 = 100 + 100 = \boxed{200}$.

Verified answers

Estate 500, claims $(400, 300) \to (300, 200)$. Estate 100, claims $(50, 100) \to (25, 75)$. (Proportional rule, if asked instead: $x_i = \frac{A}{C} c_i$.)

CHAPTER 10

Family 6 — First-Price Auction

Frequently Asked PYQ Concept

n bidders, private values uniform on $[0,1]$, symmetric equilibrium. Optimal bid and expected revenue.

Bid $b(v) = \dfrac{n-1}{n}v$; expected revenue $\dfrac{n-1}{n+1}$ (the expected second-highest value).

Verified answers

Five bidders ($n=5$, bid $\frac{4}{5}v$): $v=0.75 \to 0.6$; $v=0.6 \to 0.48$; expected revenue $\frac{4}{6}=\frac{2}{3}$.
Six bidders ($n=6$, bid $\frac{5}{6}v$): $v=0.9 \to 0.75$; expected revenue $\frac{5}{7}$.

Quiz Trap

Bid fraction $\frac{n-1}{n}$; revenue fraction $\frac{n-1}{n+1}$ — different. Revenue equivalence: the second-price/English auction gives the same expected revenue.

CHAPTER 11

Family 7 — Sponsored Search / VCG

Frequently Asked PYQ Concept

Two ad slots, clickthrough rates 10 (slot a) and 5 (slot b); three advertisers x, y, z valuing a click at $3, 2, 1$. Find the socially optimal total value and the VCG prices.

Efficient assignment: highest value to highest clickthrough: $x \to a$, $y \to b$, z unassigned. Total value $= 3(10) + 2(5) = 40$. **VCG (externality) prices:**

$$\text{slot } a: \quad \underbrace{[2(10) + 1(5)]}_{\text{others without } x=25} - \underbrace{[2(5)]}_{\text{others with } x=10} = 15, \qquad \text{slot } b: \ [3(10) + 1(5)] - [3(10)] = 5.$$

Verified answers

Socially optimal total $=$ **40**; VCG price slot $a =$ **15**, slot $b =$ **5**. (Per-click: 1.5 and 1.) Truthful bidding is dominant; the single-slot case is the second-price auction.

CHAPTER 12

Family 8 — Order Statistics

Frequently Asked PYQ Concept

"Five numbers drawn independently from Uniform$[0, 1]$"; judge statements about expected maximum / minimum / order positions.

Formula Insight

$\mathbb{E}[k\text{-th smallest of } n] = \dfrac{k}{n+1}$. For $n = 5$: $\mathbb{E}[\text{min}] = \frac{1}{6}$, $\mathbb{E}[\text{2nd}] = \frac{2}{6} = \frac{1}{3}$, median $\frac{3}{6} = \frac{1}{2}$, $\mathbb{E}[\text{2nd-highest}] = \frac{4}{6} = \frac{2}{3}$, $\mathbb{E}[\text{max}] = \frac{5}{6}$.

★ Exam Tip

These power the auction revenue ($\mathbb{E}[\text{2nd-highest}]$). Evaluate each statement against $\frac{k}{n+1}$ and pick the matching alternative.

CHAPTER 13

Family 9 — Bayes' Rule

Frequently Asked PYQ Concept

Two flavours: a *spam filter* and a *defective-item/machine* problem; each asks a total probability and a posterior.

13.1 Spam filter

40% spam; $\Pr[\text{phrase} \mid \text{spam}] = 1\%$, $\Pr[\text{phrase} \mid \text{ham}] = 0.4\%$.

$$\Pr[\text{phrase}] = 0.4(0.01) + 0.6(0.004) = 0.0064, \quad \Pr[\text{spam} \mid \text{phrase}] = \frac{0.004}{0.0064} = 0.625.$$

13.2 Defective item (two machines)

Machine A makes 25% (defect rate 6%), machine B makes 75% (defect rate 3%):

$$\Pr[\text{def}] = 0.25(0.06) + 0.75(0.03) = 0.0375, \quad \Pr[A \mid \text{def}] = \frac{0.015}{0.0375} = 0.40.$$

Verified answers

Spam: total Pr $= 0.0064$, posterior $\Pr[\text{spam} \mid \text{phrase}] = 0.625$. Defective: $\Pr[\text{def}] \approx 0.0375$, $\Pr[A \mid \text{def}] = 0.40$. Always divide by the *total* probability of the evidence.

CHAPTER 14

Family 10 — The Campaign Repeated Game

Frequently Asked PYQ Concept

Two parties choose "negative-campaign" levels a_1, a_2; reaction function $\mathrm{BR}_i(a_j) = \frac{1+a_j}{2}$. The one-shot game is a Prisoner's-Dilemma-like conflict; then it is repeated infinitely with a binding "$a_1 = a_2 = 0$" agreement.

Verified answers

Best response to $a_1 = 1.5$: $\mathrm{BR}_2 = \frac{1+1.5}{2} = \mathbf{1.25}$. **Unique symmetric NE** campaign level: $a^* = \frac{1+a^*}{2} \Rightarrow a^* = \mathbf{1}$. **Equilibrium (stage-NE) payoff:** $\mathbf{-1}$ each. **Cooperative** stage payoff ($a = 0$): $\mathbf{0}$ each. **Best deviation** from cooperation: campaign level $\frac{1}{2}$, yielding $\frac{1}{4}$. **Sustainable** by grim trigger iff $\delta \geq \frac{1}{5}$, so the agreement *cannot* be a SPE iff $\boxed{\delta \leq \frac{1}{5}}$.

The $\delta \geq \frac{1}{5}$ threshold

One-shot deviation gains $T - R = \frac{1}{4} - 0 = \frac{1}{4}$; punishment reverts to the stage NE payoff -1 forever, losing $R - P = 0 - (-1) = 1$ per future period. Cooperate iff $\frac{1}{4} \leq \frac{\delta}{1-\delta} \cdot 1 \Rightarrow \frac{1}{4}(1-\delta) \leq \delta \Rightarrow \delta \geq \frac{1}{5}$.

CHAPTER 15

Family 11 — Roommate / Effort Best-Response

Frequently Asked PYQ Concept

Two players choose effort/cleaning time; payoffs make each one's effort *less* valuable when the other does more. Find a best response and the symmetric pure Nash equilibrium.

Verified answers

Best response of roommate 2 when roommate 1 chooses $t_1 = 0$: **5** (the unconstrained optimum of a single roommate). In the effort version the unique symmetric pure NE is $e_1 = e_2 = \mathbf{a}$ (the standalone coefficient; keyed so the relevant value = 10). Method: set $\partial u_i/\partial t_i = 0$ to get $\mathrm{BR}_i(t_j)$, then impose symmetry $t_i = t_j$ — a Cournot-style fixed point.

Quiz Trap

This is a *continuous* game: differentiate the payoff, set to zero, solve the reaction functions simultaneously (exactly the Cournot technique from Book 1).

CHAPTER 16

Family 12 — Weak Dominance Killing a Nash Equilibrium

Frequently Asked PYQ Concept

"P: (D, R) is a Nash equilibrium. Q: eliminating weakly dominated strategies also eliminates (D, R), a Nash equilibrium. Choose the correct alternative."

Verified answer

Both P and Q are true. A profile can be a genuine Nash equilibrium yet still be removed by *iterated elimination of weakly dominated strategies* — because weak elimination is *not* equilibrium-preserving (unlike strict elimination). This is the Book 1 IEWDS caution made into a true/false item.

CHAPTER 17

Family 13 — Gale–Shapley Matching

Frequently Asked PYQ Concept

"Mr. a/b/c/d will be paired with ..." and statements defining *stability*.

Verified answers

Run deferred acceptance for the stated proposing side (mechanics in Book 2). A matching is **stable** iff no *blocking pair* exists — no unmatched man–woman couple who both prefer each other to their assigned partners. The keyed "correct alternative" is the one stating precisely this (an unmatched pair would deviate only if *both* strictly prefer each other).

CHAPTER 18

Family 14 — Mixed-Entry / Declining-Industry Games

Frequently Asked PYQ Concept

(a) Three symmetric players each "enter" with probability p; find p. (b) Two firms in a declining industry, each with three exit/stay choices; find the Nash equilibria.

(a) Symmetric mixed entry. Make a player indifferent between entering and staying out, given the *others* mix with probability p; solve the indifference equation for p (same technique as the drug-dealer/bystander games in Book 1). **(b) Declining industry (war of attrition / exit game).** Build the 3×3 matrix and use the underline method; typically there are asymmetric pure equilibria (one firm exits) plus a mixed equilibrium — the structure of a *war of attrition*.

★ Exam Tip

For any "symmetric mixed equilibrium, find p" question: write *expected payoff(enter)* = *payoff(stay out)* as a function of the others' p, then solve. Don't forget the binomial weighting when there are ≥ 3 players.

CHAPTER 19

Cumulative & Out-of-Scope Items

Cumulative game theory

The papers also reuse: **Nash equilibria / dominant strategies / backward induction** (Book 1), **cooperative-game coalition-form "watch/glove" markets** (Shapley/-core, Week 9), and the order-statistics/auction links above. Methods are unchanged.

Scope note — please verify with the instructor

Some sittings bundle **design-thinking** items ("Design Thinking for Data-Driven Apps," GP1/GP2/GP3 sprints, free-pizza incentives) and **credit-risk analytics** items ("Roll-Forward rate, DPD, MOB"). These are *not* game theory and do not appear in the Weeks 9–12 slides — they are cross-listed from other modules sharing the exam sitting. Treat the design-thinking ones with the empathize → define → ideate → prototype → test logic (Book 2, Family 7); the analytics items require that module's formulae. *Please confirm these with the course instructor.*

Part III

Concepts Found Only in the PYQs

CHAPTER 20

Gap-Filling Tools

20.1 Order-Based Shapley by Hand

Formula Insight

$n = 3$: $\phi_i = \frac{v(i)}{3} + \frac{(v(ij)-v(j))+(v(ik)-v(k))}{6} + \frac{v(N)-v(jk)}{3}$. Weighted voting: $\phi_i = \Pr[i \text{ pivotal over random orders}]$ (Shapley–Shubik).

20.2 Order Statistics for Auctions

Formula Insight

n i.i.d. Uniform$[0,1]$: $\mathbb{E}[k\text{-th smallest}] = \frac{k}{n+1}$, so $\mathbb{E}[\text{highest}] = \frac{n}{n+1}$, $\mathbb{E}[\text{2nd}] = \frac{n-1}{n+1}$. First-price bid $\frac{n-1}{n}v$; expected revenue $\frac{n-1}{n+1}$.

20.3 Bankruptcy Rules

Formula Insight

Proportional $x_i = \frac{A}{C}c_i$. Contested-garment (2): concede $A - c_j$ to i, split the contested remainder equally. Feasibility $0 \le x_i \le c_i$, $\sum x_i = A$.

20.4 VCG Externality Pricing

Formula Insight

Assign to maximise total value; charge i the harm to others = (others' best value without i) − (others' value with i). Truthful; single slot = second price.

20.5 Network Demand

Formula Insight

$U(\theta) = a + 3\theta\nu n^e$, $\theta \sim U[0,1]$. Indifferent $\hat{\theta} = \frac{p-a}{3\nu n^e}$; fulfilled-expectations $p = a + 3\nu n(1-n)$; $p_{\max} = a + \frac{3\nu}{4}$ at $n = \frac{1}{2}$.

Part IV

Quick Revision Sheets

CHAPTER 21

Last-Minute Revision

21.1 One-Page Summary (Weeks 9–12)

Remember This

Shapley value = average marginal contribution over $n!$ orders; unique under efficiency/symmetry/null/additivity. **Power index** = pivotality; null players score 0. **Bankruptcy:** proportional vs contested-garment. **Auctions:** second-price truthful; first-price bid $\frac{n-1}{n}v$; revenue $\frac{n-1}{n+1}$; revenue equivalence; reserve prices raise revenue; winner's curse with common values. **Network effects:** $p = a+3\nu n(1-n)$ hump $\Rightarrow n = 0$/critical-mass/stable; tipping, lock-in, expectations, history. **VCG:** efficient assignment + externality pricing = truthful.

21.2 Formula Card

Formula Insight

$$\phi_i^{(n=3)} = \frac{v(i)}{3} + \frac{(v(ij)-v(j))+(v(ik)-v(k))}{6} + \frac{v(N)-v(jk)}{3}; \quad \phi_i^{\text{power}} = \Pr[\text{pivotal}].$$

Bankruptcy: $x_i = \frac{A}{C}c_i$ or contested-garment. $\quad b(v) = \frac{n-1}{n}v, \quad \text{revenue} = \frac{n-1}{n+1}$.

$$\hat{\theta} = \frac{p-a}{3\nu n^e}, \quad p = a + 3\nu n(1-n). \quad \text{VCG}_i = \text{others-without-}i - \text{others-with-}i.$$

21.3 Common Traps

Quiz Trap

1. Confusing the bid fraction $\frac{n-1}{n}$ with the revenue fraction $\frac{n-1}{n+1}$.
2. Treating voting *weight* as *power* (compute pivotality).
3. Dropping the $(1-n)$ term in the network demand curve (loses the hump/tipping).
4. Forgetting a null player scores 0 in the power index.
5. Contested-garment $\neq$ equal split (except special cases); respect $0 \leq x_i \leq c_i$.
6. VCG: charging own value instead of the *externality*.

Part V

Exam Strategy Guide

CHAPTER 22

How to Attack the End-Term

22.1 Triage

If you see...	...do this
"coalition function / $v(S)$"	Shapley by $n!$ orders or $n = 3$ formula; spot null/symmetric players.
"$[q; w_1, \dots]$ / power index"	Pivotality; null $= 0$, symmetric split equally.
"assets $<$ claims / creditors"	Proportional or contested-garment.
"uniform values / sealed bid"	Bid $\frac{n-1}{n}v$; revenue $\frac{n-1}{n+1}$.
"clickthrough / slots / ads"	Efficient assignment + VCG externality.
"fulfilled-expectations / network"	$\hat{\theta} = \frac{p-a}{3\nu n^e}$, then $p = a + 3\nu n(1-n)$.
"spam / phrase / % "	Bayes: $\frac{sX}{sX+(1-s)Y}$.
"matching / cascade / voting / Nash"	Reuse Book 1/Book 2 methods.

22.2 Time Management & High-Frequency Topics

Remember This

Bank the formula-driven items first (auction bid/revenue, network $\hat{\theta}$, bankruptcy, power index). Reserve time for multi-part Shapley/coalition comprehensions (most sub-marks). **Frequency:** network-effects demand and Shapley/power index appear in *every* paper; auctions and bankruptcy are close behind. Never leave a true/false or numeric SA blank — a verified formula settles it.

Final word

Weeks 9–12 are about *value creation and its division*: Shapley splits a fair pie, bankruptcy divides a short one, auctions and VCG price scarce goods, and network effects decide whether a market even forms. Recognise the family, apply the formula, check the equilibrium/efficiency condition, and report exactly what is asked.

End of Book 3 — End-Term Preparation Guide.
Good luck. Made by Anmol Kansal with AI.